

Chapter 6

Space Vector Analysis of PMSMs



6.1 Rotor Structure and Physical Model of PMSMs

A PMSM establishes a rotor magnetic field that synchronously rotates with the rotor through permanent magnets mounted on the rotor. The stator windings generate the armature magnetic field, which combines with the rotor magnetic field to form the resultant air gap (working) magnetic field.

The type of permanent magnet material significantly affects the performance of a PMSM. Currently, neodymium-iron-boron (NdFeB), and other permanent magnet materials are widely used in automotive motors. The rotor structure of a PMSM is classified by the installation form of the permanent magnets, primarily into surface-mounted and interior-types, as shown in Figs. 6.1 and 6.2.

Interior-type PMSMs embed or insert permanent magnets within the rotor core, enhancing structural reliability and allowing for higher operating speeds. In particular, because the permeability of permanent magnets is close to that of vacuum, the air gap of the interior-type rotor structure is non-uniform, i.e., the rotor has a “salient-pole” structure. This introduces a reluctance torque component, which improves the torque-to-current ratio. Consequently, for a given torque requirement, the required magnet flux can be reduced, allowing for smaller magnets. This benefits field-weakening operation, extends the speed range, and lowers cost (Wang et al. 2022). Therefore, interior-type PMSMs are mainly used in automotive applications.

6.1.1 Physical Model of Surface-Mounted PMSM

The following assumptions are made: the magnetic reluctance of the stator and rotor cores is ignored, and eddy current and hysteresis losses are not considered, the electrical conductivity of the permanent magnet material is zero, and the magnetic permeability inside the permanent magnets is the same as that of air.

Fig. 6.1 Surface-mounted rotor structure

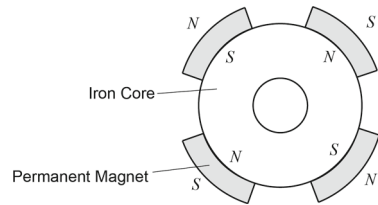
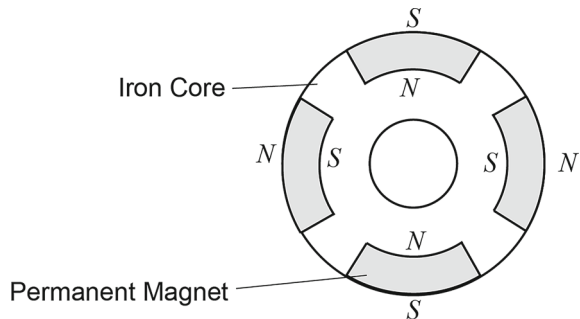


Fig. 6.2 Interior-type rotor structure



For each type of rotor structure, depending on the type of permanent magnet material and design requirements, the shape of the permanent magnet and the structural form of the rotor can vary, with various design schemes possible. However, a fundamental principle is to generate a sinusoidal distribution of the excitation magnetic field as much as possible, considering cost, manufacturing, and reliable operation. The permanent magnet excitation magnetic field can generate a sinusoidal induced EMF in the phase windings, which is the basic assumption for describing PMSMs using space vectors.

Figure 6.3 shows the physical model of a two-pole surface-mounted PMSM. The stator is modeled as an equivalent three-phase AC winding, with the positive directions of each phase's voltage and current marked, and their positive directions aligned. The voltage and current can take arbitrary waveforms and instantaneous values. The axis of the sinusoidal MMF generated by a forward current through a phase winding is defined as the axis of that phase winding. The A-axis is taken as the spatial reference axis of the ABC coordinate system. Similarly, the three-phase windings can be represented as axis coils located along the A, B, and C axes. The counterclockwise direction is defined as the positive direction for rotational speed and electromagnetic torque.

For the surface-mounted rotor structure, since the permeability inside the permanent magnets is very low, close to that of air, the permanent magnets placed on the rotor surface can be equivalent to excitation windings placed in the rotor slots. It is assumed that the sinusoidal distribution of the excitation magnetic field produced in the air gap by the excitation windings is the same as that produced by the two permanent magnets, as shown in Fig. 6.3a.

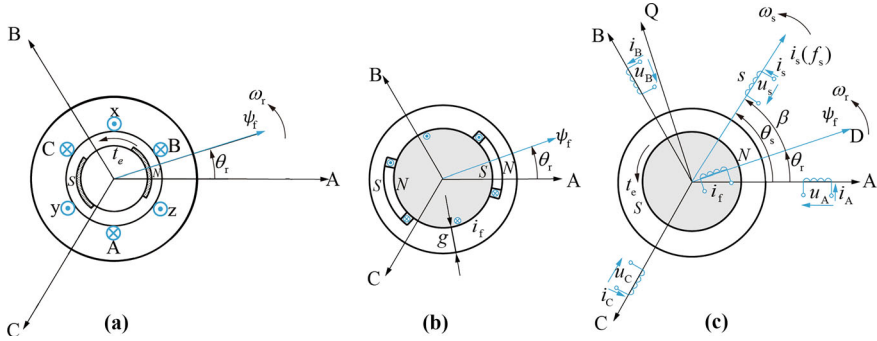


Fig. 6.3 Physical model of a two-pole surface-mounted PMSM: **a** Structural diagram; **b** PM excitation winding equivalent; **c** physical model

Assume the effective number of turns of the rotor equivalent excitation winding is $3/2$ times that of the phase winding, and after passing the equivalent excitation current i_f , the sinusoidal distribution of the excitation magnetic field produced in the air gap is the same as that of the permanent magnets. It is assumed that the equivalent excitation inductance is M_{sf} , neglecting leakage magnetic fields, and the rotor excitation flux linkage is

$$\psi_f = M_{sf} i_f \quad (6.1)$$

Considering the equivalence of turns between stator and rotor, the rotor excitation flux linkage is also equal to the mutual flux linkage of the permanent magnets generated in the stator. During PMSM operation, if the effect of temperature changes on the magnetic supply capability of the permanent magnets is not considered, it can be assumed that ψ_f is constant, i.e., i_f is a constant value. Since the permeability inside the permanent magnets is close to that of air, the air gap of the surface-mounted PMSM is uniform, with an air gap length of g . This results in the physical model of the surface-mounted PMSM shown in Fig. 6.3c.

In the physical model, the equivalent excitation winding is represented as a coil located on the axis of the permanent magnet excitation magnetic field. The axis of the permanent magnet excitation magnetic field is defined as the D axis, with the Q axis leading the D axis by an electrical angle of 90° . Represented by space vectors, the rotor flux linkage is ψ_f , f_s and i_s are the MMF vector and stator current vector generated by the stator three-phase winding, respectively. The equivalent single-axis coil generating $i_s(f_s)$ is located along the $i_s(f_s)$ axis, and its effective number of turns is $3/2$ times that of a phase winding. The angle between the stator current (MMF) vector and the D axis is β , with the counterclockwise direction taken as positive. Relative to the fixed magnetic axis of the stator phase A winding, the rotor rotates with a (mechanical) angular velocity of ω_r and a spatial angle of θ_r , and the stator current (magnetomotive force) vector rotates with an angular velocity of ω_s and a spatial angle of θ_s . Under steady-state conditions and for a single pole-pair machine,

the stator current rotation frequency equals the rotor rotation frequency, i.e., $\omega_s = \omega_r$, and

$$\beta = \theta_s - \theta_r \quad (6.2)$$

For the same stator current amplitude, β angle determines the output torque magnitude, thus it is also called the **torque angle**.

6.1.2 Physical Model of Interior-Type PMSM

Figure 6.4 shows a schematic diagram and equivalent physical model of a two-pole interior-type PMSM. Similarly, two permanent magnets of the interior-type rotor can be equivalently represented as excitation windings placed in the rotor slots, with an effective number of turns $3/2$ times that of the phase winding, and an equivalent excitation current of i_f . Unlike the surface-mounted PMSM, the air gap of the interior-type PMSM is no longer uniform. In this case, the air gap length along the axis of the permanent magnet increases to $g + h$, where h is the height of the permanent magnet, while the air gap length facing the rotor core remains g . As a result, the magnetic permeance in the D-axis direction of the rotor is lower than that in the Q-axis direction. By selecting coordinate axes and angle representations similar to the surface-mounted PMSM, the physical model of the interior-type PMSM shown in Fig. 6.4c can be obtained. Similarly, the air gap magnetic field of the interior-type PMSM is formed by the superposition of the rotor permanent magnet magnetic field and the stator armature reaction magnetic field.

As shown in Fig. 6.4c, for the interior-type rotor structure, the air gap is non-uniform. Under the same amplitude of i_s , the equivalent excitation inductance varies with the β angle. Due to different spatial phase angles β , the resulting armature reaction magnetic field differs, which makes it difficult to quantitatively calculate and

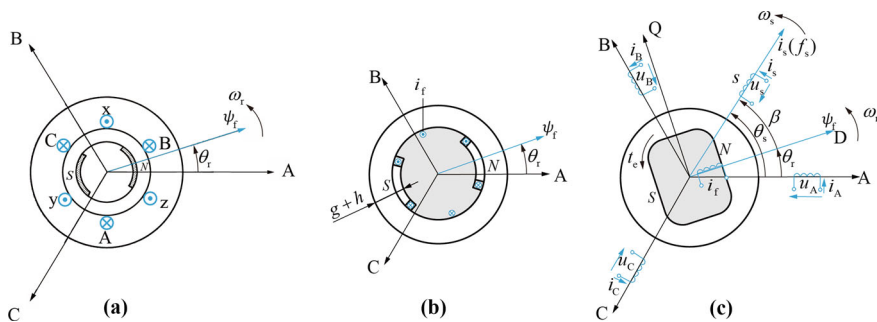


Fig. 6.4 Schematic diagram and equivalent physical model of two-pole interior-type PMSM: **a** Schematic diagram; **b** rotor equivalent excitation winding; **c** physical model

analyze the armature reaction. In electrical machine theory, the **two-reaction theory** is commonly adopted to analyze interior-type PMSM. For this purpose, the DQ synchronous reference frame shown in Fig. 6.4c is used to construct the mathematical model. When $\beta = 0^\circ$, the sinusoidal distribution magnetic field generated by $i_s(f_s)$ in the air gap is called the direct-axis armature reaction magnetic field, denoted as D; when $\beta = 90^\circ$, the sinusoidal distribution magnetic field generated by $i_s(f_s)$ in the air gap is called the quadrature-axis armature reaction magnetic field, denoted as Q. Since the direct-axis magnetic permeance is lower than the quadrature-axis magnetic permeance, under the same amplitude of $i_s(f_s)$, the direct-axis armature reaction magnetic field is weaker than the quadrature-axis armature reaction magnetic field, i.e., $L_{mD} < L_{mQ}$, where L_{mD} and L_{mQ} are the direct-axis equivalent excitation inductance and quadrature-axis equivalent excitation inductance, respectively, also referred to as **direct-axis inductance** and **quadrature-axis inductance**.

For surface-mounted PMSM, there is $L_{mD} = L_{mQ} = M_{sf}$, which is collectively referred to as **equivalent excitation inductance**.

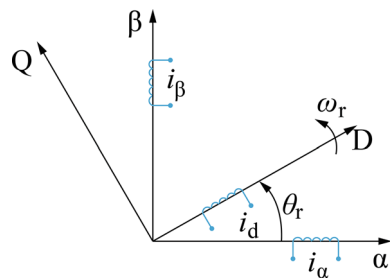
6.1.3 Equivalent Primitive Machine of the PMSM Physical Model

Whether for surface-mounted or interior-type PMSM, when using the primitive machine model for analysis, the orientation of the rotor's orthogonal equivalent d- and q-axis windings should be explicitly defined: the d-axis winding aligns with the permanent magnet flux axis, since the permanent magnets are arranged along the D-axis; the q-axis winding is in an open-circuit state and can be omitted.

The equivalent physical model of the four-coil primitive machine is shown in Fig. 6.5.

Under the excitation of permanent magnet, it is assumed that the equivalent excitation current is i_f , i.e., assuming rotor current $i_d = i_f$, and $L_{Dd} = M_{sf}$, then the permanent magnet mutual flux linkage is $\psi_f = M_{sf}i_f = L_{Dd}i_f$.

Fig. 6.5 The equivalent four-coil primitive machine model of the PMSM



6.2 Stator Flux Linkage and Voltage Equations

6.2.1 Stator Flux Linkage Vector

The rotor structure of an interior PMSM is salient-pole, resulting in a non-uniform air gap. The excitation effect of the current vector varies with its spatial orientation, leading to salient-pole effects. As previously mentioned, this can be addressed using either the two-reaction theory or the complex inductance method.

The two-axis theory is the most representative method for analyzing salient-pole machines. Based on the aforementioned equivalent primitive machine model of the PMSM, the rotor-fixed DQ coordinate system is introduced. The rotor of the PMSM consists of permanent magnets, and the rotor-fixed synchronous DQ coordinate system is aligned with the magnetic field generated by the permanent magnets. Under the equivalent primitive machine model of the PMSM, the rotor's q-axis excitation current is zero, and only the D-axis flux linkage component of the stator contains the mutual flux linkage from the permanent magnets, denoted as ψ_f . In the DQ coordinate system, the stator current and flux linkage vectors are expressed by their coordinate components as $\mathbf{i}_s^D = i_D + j i_Q$, $\boldsymbol{\psi}_s^D = \psi_D + j \psi_Q$, where i_D and i_Q are the stator current components, and ψ_D and ψ_Q are the stator flux linkage components in the DQ coordinate system.

According to the stator flux linkage equations of the four-coil primitive machine in Chap. 4, the flux linkage expressions of the PMSM in the rotor-fixed DQ synchronous coordinate system can be written as:

$$\psi_D = L_D i_D + \psi_f \quad (6.3)$$

$$\psi_Q = L_Q i_Q \quad (6.4)$$

where L_D is the direct-axis synchronous inductance, $L_D = L_{s\sigma} + L_{mD}$, L_Q is the quadrature-axis synchronous inductance, $L_Q = L_{s\sigma} + L_{mQ}$, $L_{s\sigma}$ is the leakage inductance of the phase winding, and $\psi_f = M_{sf} i_f$ is the permanent magnet mutual flux linkage. If the PMSM is of the interior-type, the direct-axis and quadrature-axis synchronous inductances are unequal. Since the permanent magnets are mounted along the D-axis, the magnetic reluctance in this direction is higher, resulting in $L_D < L_Q$. If the PMSM is of the surface-mounted type, then $L_D = L_Q = L_s$.

Additionally, complex inductance can be introduced to handle the salient-pole effects of interior-type PMSM, and the stator flux linkage equation can be expressed in vector form,

$$\boldsymbol{\psi}_s^D = \mathbf{L}_s \mathbf{i}_r^D + \boldsymbol{\psi}_f^D \quad (6.5)$$

where L_s is the complex inductance, with its derivation detailed in Appendix B.1. If the PMSM is surface-mounted, the complex inductance L_s is a real number. Equation (6.5) indicates that in the presence of rotor salient-pole effects, the directions of the stator self-inductance flux linkage vector and the excitation current vector are not aligned, and the complex inductance reflects this misalignment.

By introducing complex inductance, and based on vector transformation, both sides of Eq. (6.5) can be multiplied by $e^{j\theta_r}$, yielding the flux linkage vector equation of the PMSM in the stationary $\alpha\beta$ coordinate system

$$\psi_s = L_s i_s + \psi_f \quad (6.6)$$

The magnetic field generated by the stator current vector (the sum of the leakage field and the armature reaction field) is usually referred to as the armature field. The excitation field of the rotor permanent magnet is called the rotor field. It can be observed that $L_s i_s$ is the armature flux linkage vector, corresponding to the armature magnetic field; the rotor flux linkage vector ψ_f represents the rotor magnetic field linked by the stator windings (mutual flux linkage); and the stator magnetic field is the resultant field composed of the armature and rotor magnetic fields. The stator flux linkage vector ψ_s represents the total flux linkage of the stator.

The space vector of the permanent magnet mutual flux linkage linked by the three-phase symmetrical stator windings can be regarded as the resultant of the mutual flux linkages of the three-phase windings

$$\psi_f = \frac{2}{3}(\psi_{fA} + a\psi_{fB} + a^2\psi_{fC}) \quad (6.7)$$

where ψ_{fA} , ψ_{fB} , ψ_{fC} represent the permanent magnet mutual flux linkages of the A, B, and C windings, respectively. The vector $\psi_f = \psi_f e^{j\theta_r}$ is a rotating vector, with a rotational angular velocity of ω_r , and θ_r denotes the rotor electrical angle, i.e., the spatial phase of ψ_f within the ABC coordinate system, as shown in Fig. 6.4c.

6.2.2 Stator Voltage Equation

In Sect. 4.1 of Chap. 4, the concepts of the non-Faraday reference frame and the EEMF are introduced. These concepts will now be applied to derive the voltage equations of a PMSM under different coordinate systems.

For the stator windings, the stationary $\alpha\beta$ coordinate system serves as the Faraday reference frame. Therefore, by directly applying Faraday's law of electromagnetic induction, the stator voltage equation is given as follows,

$$u_s = R_s i_s + \frac{d\psi_s}{dt} \quad (6.8)$$

The DQ synchronous coordinate system rotates relative to the stator windings at an angular velocity of ω_r , resulting in relative motion and making it a non-Faraday

reference frame. Therefore, the EEMF must be added to the left side of the voltage equation, yielding

$$\mathbf{u}_s^D + \mathbf{e}_{ms}^D = R_s \mathbf{i}_s^D + \frac{d\boldsymbol{\psi}_s^D}{dt} \quad (6.9)$$

where the EEMF reflects the motional EMF induced by the rotation of the reference coordinate system with.

$$\mathbf{e}_{ms}^D = -j\omega_r \boldsymbol{\psi}_s^D \quad (6.10)$$

Under steady-state conditions, $\boldsymbol{\psi}_s^D$ does not vary with time, and the voltage equation becomes

$$\mathbf{u}_s^D + \mathbf{e}_{ms}^D = R_s \mathbf{i}_s^D \quad (6.11)$$

After the rotational speed increases, the voltage drop caused by the winding resistance becomes relatively small and can be neglected, i.e.,

$$\mathbf{u}_s^D + \mathbf{e}_{ms}^D \approx 0 \quad (6.12)$$

At this time, the stator terminal voltage is mainly used to balance the back EMF caused by the coupling motion, in order to maintain the continuous rotation of the stator (total) flux linkage. Substituting Eq. (6.10) into the flux linkage expression gives:

$$\mathbf{u}_s^D \approx j\omega_r \boldsymbol{\psi}_s^D \quad (6.13)$$

6.2.3 Decomposition of the Voltage Vector Equation in the Synchronous Coordinate System

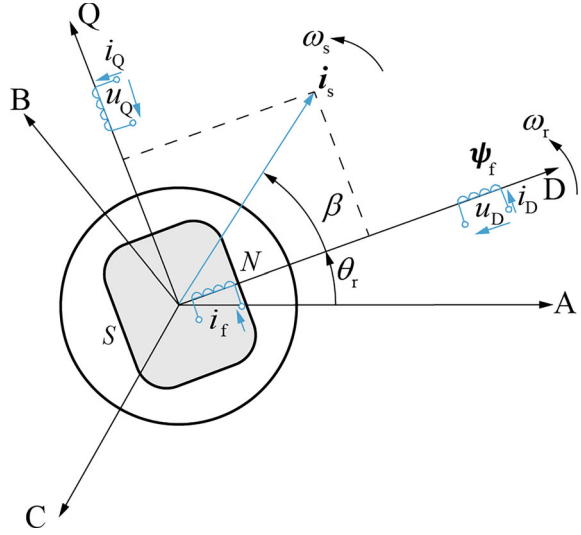
As shown in Fig. 6.6, in the DQ coordinate system, the stator current vector $\mathbf{i}_s^D$ is decomposed into:

$$\mathbf{i}_s^D = i_D + ji_Q \quad (6.14)$$

Substituting the flux linkage expression (6.5), the voltage vector equation in the DQ synchronous reference frame can be further expressed as:

$$\mathbf{u}_s^D = R_s \mathbf{i}_s^D + L_s \frac{d\mathbf{i}_s^D}{dt} + j\omega_r (L_s \mathbf{i}_s^D + \boldsymbol{\psi}_f^D) \quad (6.15)$$

Fig. 6.6 Synchronously rotating DQ coordinate system



By projecting the stator voltage Eq. (6.9) onto the D and Q axes, the voltage component equations are obtained as:

$$\begin{cases} u_D = R_s i_D + \frac{d\psi_D}{dt} - \omega_r \psi_Q \\ u_Q = R_s i_Q + \frac{d\psi_Q}{dt} + \omega_r \psi_D \end{cases} \quad (6.16)$$

The second term on the right-hand side of the equation represents the transformer EMF induced by the variation of the stator flux component, while the third term is the motional EMF. It is important to note that the signs of the motional EMF in the two equations are different. The sign of the motional EMF can be determined as follows: let the coil generating the motional EMF rotate 90° electrical degrees in the direction of rotation. If its axis aligns with the axis of the magnetic field producing the EMF, the motional EMF takes a positive sign in the equation; otherwise, it takes a negative sign. For example, after ψ_Q rotating 90° electrical degrees counterclockwise, its axis opposes the direction of u_D , so the $\omega_r \psi_Q$ term in the u_D voltage equation should have a negative sign. Similarly, after ψ_D rotating 90° electrical degrees counterclockwise, its axis aligns with the direction of u_Q , so the $\omega_r \psi_D$ term in the u_Q voltage equation should have a positive sign.

Substituting the flux linkage expressions (6.3) and (6.4), Eq. (6.16) can be expressed as

$$\begin{cases} u_D = R_s i_D + L_D \frac{di_D}{dt} - \omega_r L_Q i_Q \\ u_Q = R_s i_Q + L_Q \frac{di_Q}{dt} + \omega_r L_D i_D + e_{mf} \end{cases} \quad (6.17)$$

where $e_{mf} = \omega_r \psi_f$ is the component of the motional EMF induced by the permanent magnet flux linkage in the stator windings. It is sometimes referred to as the back EMF of the permanent magnet, and it is independent of the stator excitation current.

Under sinusoidal steady-state conditions with $di_D/dt = 0$ and $di_Q/dt = 0$, Eq. (6.17) can be expressed as

$$\begin{cases} u_D = R_s i_D - \omega_r L_Q i_Q \\ u_Q = R_s i_Q + \omega_r L_D i_D + e_{mf} \end{cases} \quad (6.18)$$

From Eq. (6.18), the steady-state vector diagram of an interior-type PMSM can be obtained, as shown in Fig. 6.7. It can be seen that due to the asymmetry of the quadrature and direct axis magnetic circuits, the stator current vector is decomposed into quadrature axis component and direct axis component, reflecting the analysis method of the two-reaction theory.

For the voltage equations of the above interior-type PMSM, if $L_D = L_Q = L_s$ is set, it can be transformed into the voltage equations of a surface-mounted PMSM.

Under steady-state conditions, i_s^D will not change, so the voltage equation becomes

$$u_s^D = R_s i_s^D + j\omega_s L_s i_s^D + j e_{mf} \quad (6.19)$$

From Eq. (6.19), the steady-state vector diagram of a surface-mounted PMSM can be obtained as shown in Fig. 6.8.

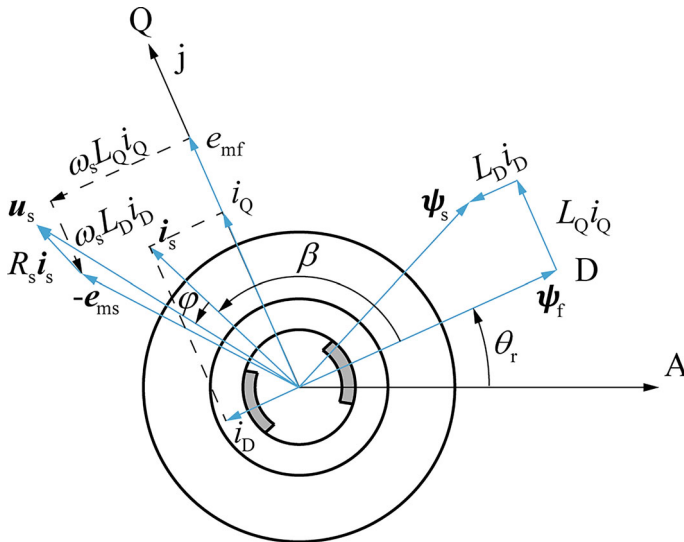


Fig. 6.7 Steady-state vector diagram of an interior-type PMSM ($i_D < 0$)

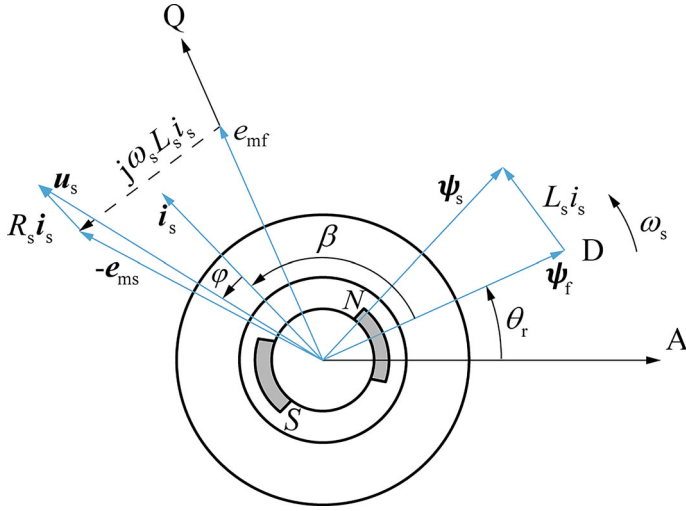


Fig. 6.8 Steady-state vector diagram of a surface-mounted PMSM

6.3 PMSM Torque Equation

6.3.1 Torque Equation

In Sect. 4.2.5, a general torque vector expression for the four-coil primitive machine is presented, and in Sect. 5.6, the inductance matrix transformation and magnetic co-energy calculation for three-phase machines are introduced. For a PMSM, the equivalent primitive machine has no rotor q-axis current component, and the d-axis current corresponds to the equivalent excitation current i_f of the permanent magnet. Therefore, the torque equation can be expressed as:

$$t_e = \frac{3}{2} p_0 (\psi_s^D \times i_s^D) \quad (6.20)$$

By substituting Eqs. (6.3), (6.4), and (6.14) into Eq. (6.20), the torque equation for a PMSM can be obtained

$$t_e = \frac{3}{2} p_0 [\psi_f i_Q + (L_D - L_Q) i_D i_Q] \quad (6.21)$$

where $\frac{3}{2} p_0 \psi_f i_Q$ is the excitation torque component, and $\frac{3}{2} p_0 (L_D - L_Q) i_D i_Q$ is the reluctance torque component. Let I_s be the amplitude of the stator current

$$I_s^2 = i_D^2 + i_Q^2 \quad (6.22)$$

In the DQ rotor-fixed synchronous reference frame oriented along the permanent magnet excitation flux, let β be the angle between the stator current vector i_s^D and the D-axis, i.e., the torque angle. In the DQ coordinate system shown in Fig. 6.4, there is

$$i_D = I_s \cos \beta \quad (6.23)$$

$$i_Q = I_s \sin \beta \quad (6.24)$$

The torque equation can be further written in polar coordinate form

$$t_e = \frac{3}{2} p_0 [\psi_f I_s \sin \beta + \frac{1}{2} (L_D - L_Q) I_s^2 \sin 2\beta] \quad (6.25)$$

when $\beta < \pi/2$, the reluctance torque is negative, acting as a braking torque,

when $\beta > \pi/2$, the reluctance torque is positive, acting as a driving torque.

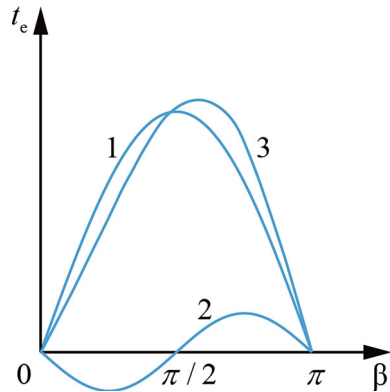
In motor drive systems based on interior-type PMSM, the reluctance torque can be utilized flexibly and effectively. For example, in the constant torque region, controlling the angle β within the range $\pi/2 < \beta < \pi$ can enhance the torque output. In the constant power region, adjusting and controlling β allows for increased output torque and an extended speed range.

It should be noted here that the reluctance torque in Eq. (6.25) is a function of the torque angle β , whereas the reluctance torque in Eq. (3.74) takes the rotor position θ_r as the independent variable. Therefore, the two expressions differ by a negative sign in form, but their fundamental variation patterns are consistent (Fig. 6.9).

For surface-mounted PMSM, it has $L_D = L_Q$, where the reluctance torque component in the torque equation is zero, i.e.,

$$t_e = \frac{3}{2} p_0 \psi_f i_Q \quad (6.26)$$

Fig. 6.9 Interior-type PMSM $t_e - \beta$ characteristic curves (1-electromagnetic torque, 2-reluctance torque, 3-composite torque)



Equation (6.27) indicates that the electromagnetic torque is determined by the Q-axis component of the stator current, i_Q referred to as the torque current.

The rotor excitation of a PMSM is not adjustable. However, the MMF vector f_s can be utilized to exert a demagnetizing effect on the permanent magnets, which is the basic principle of field-weakening control. As shown in Fig. 6.4c, when the control angle $\beta > 90^\circ$, f_s produces a direct-axis demagnetizing component f_D . For the demagnetizing MMF f_D , its effect in a surface-mounted PMSM is shown in Fig. 6.8, where the actual direction of i_D is opposite to the positive direction of the coordinate axis, i.e., $i_D < 0$.

As long as the D-axis current is not zero, there is a field-weakening effect. However, in the constant torque region and the constant power region, the purpose of applying D-axis current is different. The former is to fully utilize the reluctance torque, while the latter is to expand the operating speed range of the PMSM. Generally, only the control of the D-axis current in the constant-power region is referred to as **field-weakening control**.

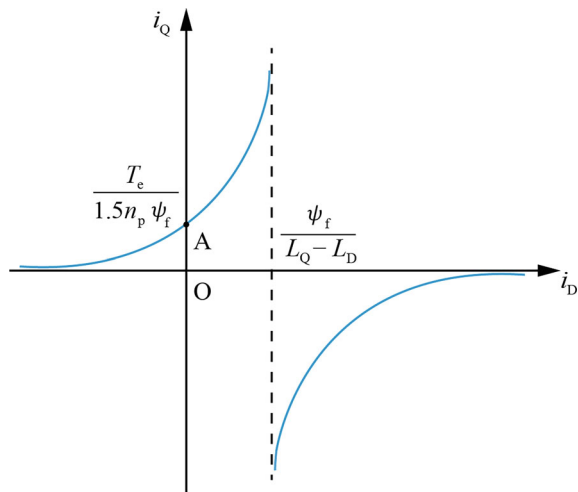
For a desired non-zero torque t_e , the relationship between i_Q and i_D currents can be obtained as

$$i_Q = \frac{2t_e}{3p_0[\psi_f + (L_D - L_Q)i_D]} \quad (6.27)$$

This is the current expression under the constant torque, which forms a hyperbola in the “current phase plane” defined by the quadrature and direct axis currents as orthogonal coordinates. The asymptotes of the hyperbola are given by $i_D = \psi_f / (L_Q - L_D)$ and $i_Q = 0$, respectively (Fig. 6.10).

From the above constant torque curve, it can be seen that to achieve the same desired torque, the stator current can be selected arbitrarily along the constant torque

Fig. 6.10 Constant torque characteristic curve in the current phase plane



curve. This gives rise to the issue of determining an appropriate “current allocation strategy” for selecting the stator current.

6.3.2 Constant Torque Curve and MTPA

Once the PMSM structure is determined, the electromagnetic torque magnitude is governed by the two components of the stator current. For each torque value t_e , there exist countless combinations of i_Q and i_D to achieve same torque t_e . Therefore, it becomes necessary to define an allocation principle for these two current components, which is known as the optimal stator current allocation strategy. Clearly, different optimization objectives will lead to different principles and control methods for allocating the two current components.

When the PMSM operates in the low-speed region, iron losses are not significant, while copper losses account for a larger proportion. Therefore, for a specific torque demand, the stator current is usually controlled according to the principle of maximizing the torque/current ratio. This not only minimizes the copper losses of the PMSM, but also reduces the losses of the motor controller, thereby lowering the system's total losses.

As described in Sect. 6.3.1, the constant torque curves form a family of hyperbolas with the asymptote $i_D = \psi_f / (L_D - L_Q)$, as indicated by the dashed lines in Fig. 6.11.

The optimization problem of stator current control is to select the optimal current combination from the constant torque curve under a given torque condition. Each constant torque curve has a point closest to the origin, which corresponds to the minimum stator current. Connecting these points across the constant torque curves forms the trajectory of the minimum stator current vector, known as the maximum torque per ampere (MTPA) curve (Vaez-Zadeh 2018), as shown by the solid line in Fig. 6.11.

From a geometric perspective, the MTPA curve is formed by connecting the tangency points between the constant torque curves and the current circles.

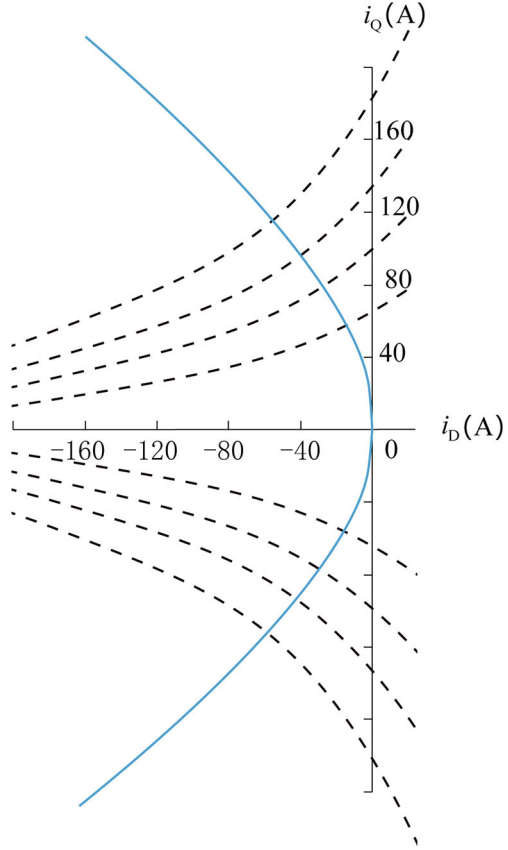
The analytical solution of the MTPA curve can be obtained using the polar form of the torque expression. For a given current amplitude I_s , setting $\partial t_e / \partial \beta = 0$ allows the solution of the MTPA. Of course, as presented in Sect. 4.3.2, the MTPA derivation based on rotor current orientation is also provided, and its result is directly referenced here:

$$i_Q = \pm \sqrt{i_D \left(i_D + \frac{\psi_f}{L_D - L_Q} \right)}, i_D < 0 \quad (6.28)$$

Alternatively, the extremum of the torque expression can be derived to obtain the solution,

$$\frac{\partial t_e}{\partial \beta} = \frac{3}{2} [\psi_f I_s \cos \beta + L_D - L_Q I_s^2 \cos 2\beta] = 0 \quad (6.29)$$

Fig. 6.11 Stator current trajectory for maximum torque/current ratio



Considering

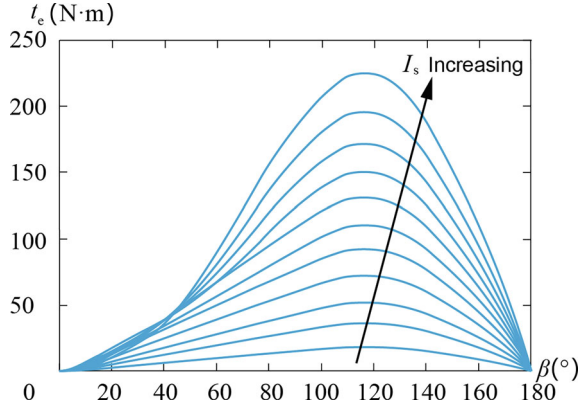
$$\begin{cases} I_s \cos \beta = i_D \\ I_s^2 \cos 2\beta = i_D^2 - i_Q^2 \end{cases} \quad (6.30)$$

the MTPA result can be obtained as

$$i_Q = \pm \sqrt{i_D \left(i_D + \frac{\psi_f}{L_D - L_Q} \right)}, \quad i_D < 0 \quad (6.31)$$

As shown in Fig. 6.11, the stator current vector trajectory is symmetrically distributed in the second and third quadrants. In the second quadrant, the torque is positive (driving mode), while in the third quadrant, the torque is negative (braking mode). The trajectory is tangent to the Q-axis at the origin, and its asymptote in the second quadrant is a 45° straight line. When the torque is relatively low, the trajectory is close to the Q-axis, indicating that the torque is mainly generated by magnetic

Fig. 6.12 Torque angle characteristics at different current magnitudes



excitation. As the torque increases, the trajectory gradually deviates from the Q-axis, implying that the contribution of reluctance torque becomes more significant. This illustrates that one of the key advantages of the interior-type PMSM is the effective utilization of reluctance torque to enhance torque generation capability. However, this comes at the cost of increased PMSM manufacturing complexity, as the rotor structure is relatively more intricate.

Figure 6.12 shows the “torque–angle” characteristics under different current magnitudes, that is, the variation of electromagnetic torque with respect to the torque angle β . The peak of each curve corresponds to the point where $\partial t_e / \partial \beta = 0$.

In engineering implementation, i_D and i_Q are generally expressed as functions of the desired torque t_e , so that the target current setpoints can be directly obtained based on the torque reference. This can be achieved either by lookup tables or by analytical solutions. The torque Eq. (6.21) can be transformed into

$$i_Q = \frac{2t_e}{3p_0[\psi_f + (L_D - L_Q)i_D]} \quad (6.32)$$

By combining this equation with MTPA Eq. (6.31), the current expressions $i_D^*(t_e)$ and $i_Q^*(t_e)$ can be achieved.

In practice, the output current capability of the inverter is limited by its rated capacity. Therefore, the stator current is constrained by a maximum allowable value I_{sm} , i.e.,

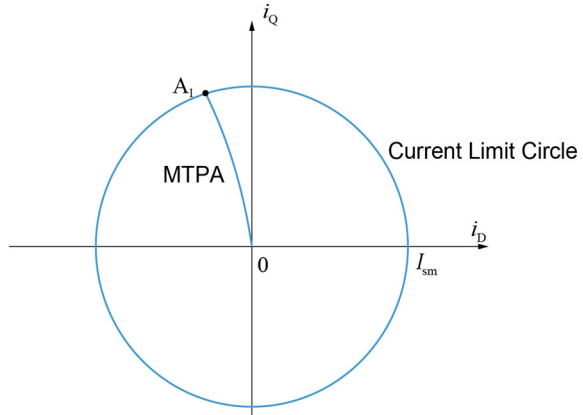
$$|i_s| \leq I_{sm} \quad (6.33)$$

If expressed in terms of the two components of the stator current vector, then

$$i_D^2 + i_Q^2 \leq I_{sm}^2 \quad (6.34)$$

Equation (6.34) forms the current limit circle, as shown in Fig. 6.13, where the radius of the current limit circle is I_{sm} . It should be noted that this expression holds

Fig. 6.13 Current limit circle and MTPA trajectory



under the assumption of the equal-amplitude coordination transformation. Under this condition, the peak value of the inverter output (phase) current equals the magnitude of the stator current vector.

6.4 Principles of FOC for PMSMs

Field-oriented control (FOC) is the predominant strategy for high-performance control of PMSMs. The stator current vector is oriented by the rotor magnetic field, FOC decouples the torque and field- weakening components of the stator current. In the low-speed region, the stator current vector is typically optimized according to the MTPA strategy to maximize torque output while minimizing copper losses. As the motor speed increases and approaches the voltage limit imposed by the inverter, field-weakening control is employed by injecting negative D-axis current to reduce the effect of the back EMF, allowing PMSMs to operate beyond its base speed while maintaining voltage and current constraints.

6.4.1 Voltage Limit Ellipse and Transition Speed

The maximum voltage that a vehicle motor controller can supply to the motor is limited by the DC voltage available from the onboard power supply. The magnitude of the stator voltage vector $\mathbf{u}_s$ required for steady-state PMSM operation is directly related to the rotor electrical angular speed ω_r , which means that the PMSM operating speed is constrained by the inverter's voltage limit.

In the sinusoidal steady-state condition, it is known from Eq. (6.18) that the voltage component equations in the DQ coordinate system are

$$u_D = R_s i_D - \omega_r L_Q i_Q \quad (6.35)$$

$$u_Q = R_s i_Q + \omega_r L_D i_D + \omega_r \psi_f \quad (6.36)$$

when the PMSM operates at high speed, the voltage drops across the resistance in Eqs. (6.35) and (6.36) become relatively small and can be neglected, it has

$$|\mathbf{u}_s|^2 = (\omega_r \psi_f + \omega_r L_D i_D)^2 + (\omega_r L_Q i_Q)^2 \quad (6.37)$$

Considering the power supply voltage constraint, it holds

$$|\mathbf{u}_s| \leq U_{sm} \quad (6.38)$$

where U_{sm} is maximum achievable magnitude of $|\mathbf{u}_s|$.

For ease of analysis, Eq. (6.38) is converted into

$$(L_D i_D + \psi_f)^2 + (L_Q i_Q)^2 = \left(\frac{|\mathbf{u}_s|}{\omega_r}\right)^2 \leq \left(\frac{U_{sm}}{\omega_r}\right)^2 \quad (6.39)$$

Equation (6.39) forms a voltage limit ellipse that constrains i_D and i_Q , as shown in Fig. 6.14.

Define the **saliency ratio** as $\rho = L_Q/L_D$, then the ellipse equation is obtained

$$\frac{1}{\rho^2} \left(i_D + \frac{\psi_f}{L_D}\right)^2 + (i_Q)^2 \leq \left(\frac{U_{sm}}{\omega_r L_Q}\right)^2 \quad (6.40)$$

Fig. 6.14 Voltage limit ellipse

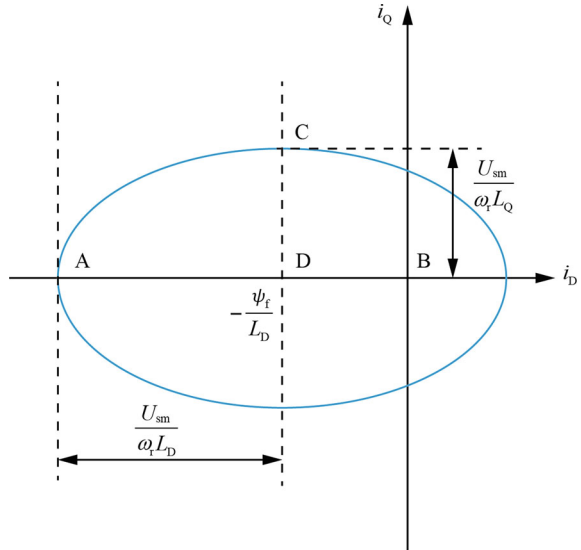
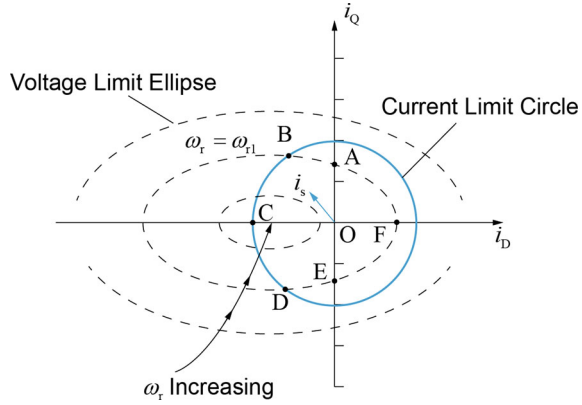


Fig. 6.15 Combined effect of voltage limit ellipse and current limit circle



The foci of the voltage limit ellipse are $(-\psi_f/L_D, 0)$, with a major axis of $U_{sm}/\omega_r L_D$ and a minor axis of $U_{sm}/\omega_r L_Q$. The lengths of the axes of the voltage limit ellipse are inversely proportional to the angular velocity ω_r . As the angular velocity increases, a series of progressively smaller concentric ellipses are formed. In the current phase plane, at low speeds (ω_r is small), the major and minor axes of the voltage ellipse are larger, imposing fewer constraints on the operating range of the PMSM; $\omega_r = 0$, the current is completely unconstrained by the voltage ellipse; as ω_r increases, the voltage ellipse shrinks, limiting the operating point of the PMSM to within the voltage limit ellipse, making the exterior of the ellipse unreachable.

The stator current vector $\mathbf{i}_s$ must satisfy both the current limit constraint equation and the voltage limit constraint equation, so the stator current vector $\mathbf{i}_s$ must fall within both the current limit circle and the voltage limit ellipse, as shown in Fig. 6.15. For example, when $\omega_r = \omega_{r1}$, $\mathbf{i}_s$ must be restricted within the range ABCDEF.

The voltage limit ellipse is a series of ellipses that shrink as the rotational speed increases. For an interior-type PMSM, in the current phase plane, as the PMSM speed increases, when the stator voltage satisfies $|\mathbf{u}_s| = U_{sm}$, i.e., $|\mathbf{u}_s|$ reaches its limit, the rotor speed at this point is defined as the transition speed, denoted as ω_{rt} .

The transition speed is the maximum speed the PMSM can reach at a given current operating point (i_D, i_Q) due to voltage constraint. Clearly, ω_{rt} is a function of the current operating point (i_D, i_Q) of the PMSM, with the expression of

$$\omega_{rt}(i_D, i_Q) = \frac{U_{sm}}{\sqrt{(L_D i_D + \psi_f)^2 + (L_Q i_Q)^2}} \quad (6.41)$$

On the MTPA curve, each operating point corresponds to a transition speed. As the current magnitude decreases, the transition speed increases; conversely, as the current magnitude increases, the transition speed decreases. The minimum transition speed occurs at the full-load condition, that is, at the intersection of the current limit circle and the MTPA curve. This speed is defined as the base speed, denoted as ω_{rb} , which is given by:

$$\omega_{rb} = \frac{U_{sm}}{\sqrt{(L_D i_{Dm} + \psi_f)^2 + (L_Q i_{Qm})^2}} \quad (6.42)$$

The maximum values of the d-axis and q-axis currents i_{Dm} and i_{Qm} satisfy the following relationship

$$\begin{cases} i_{Dm}^2 + i_{Qm}^2 = I_{sm}^2 \\ i_{Qm} = \sqrt{i_{Dm}^2 + i_{Qm}^2} \left(i_{Dm} + \frac{\psi_f}{L_D - L_Q} \right) \end{cases} \quad (6.43)$$

For a surface-mounted PMSM, the MTPA curve satisfies $i_D = 0$, and Eq. (6.42) becomes

$$\omega_{rb} = \frac{U_{sm}}{\sqrt{\psi_f^2 + (L_Q i_{sm})^2}} \quad (6.44)$$

The maximum transition speed is achieved under no-load conditions. If the no-load current is ignored, from Eq. (6.37), it holds

$$\omega_r \psi_f = |\mathbf{u}_s| = e_{mf} \quad (6.45)$$

The rotor speed at which the no-load back EMF e_{mf} attains the voltage limit U_{sm} is defined as the no-load base speed, denoted as ω_{rb0} . According to Eq. (6.41), it can be expressed as:

$$\omega_{rb0} = \frac{U_{sm}}{\psi_f} \quad (6.46)$$

Below the transition speed ($\omega_r < \omega_{rt}$), the operating point is not constrained by the voltage limit ellipse, and the PMSM can operate within the current limit circle following the MTPA trajectory. The maximum value of the current (torque) is selected as the intersection of the MTPA curve and the current limit circle. Above the transition speed ($\omega_r > \omega_{rt}$), once the voltage limit ellipse intersects with the MTPA curve, if the PMSM continues to operate along the MTPA trajectory, the output torque can no longer reach the maximum allowable value (limited by the current). At this point, the operating point of the PMSM should be adjusted according to new criterions discussed in following section.

6.4.2 Field-Weakening Control

1. Field-Weakening Principle

First, consider the characteristic curve of the PMSM. On the external characteristic curve, the base speed marks the boundary between the constant torque operation region and the constant power operation region. The basic principle of operation in the

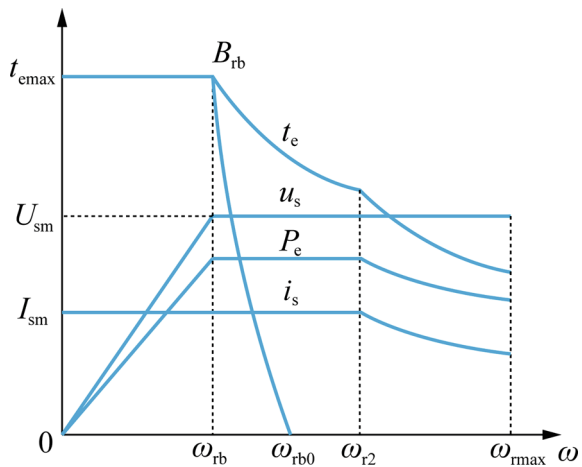
constant power region is to increase the D-axis demagnetizing current to expand the operating range of the PMSM. Therefore, the constant power operation region is also called the weak magnetic region. The line connecting the base speed operating point within the partial characteristics (within the external characteristics) $B_{rb}(\omega_{rb}, t_{e\max})$ and the no-load base speed operating point $(\omega_{rb0}, 0)$ is the weak magnetic boundary line. This section focuses on the steady-state current control rules on either side of the weak magnetic boundary line.

Figure 6.17 shows not only the voltage limit ellipse and the current limit circle but also the MTPA trajectory. For the surface-mounted PMSMs, this trajectory coincides with the Q-axis; for the interior-type PMSM, the trajectory corresponds to the stator current MTPA curve shown in Fig. 6.11. Both trajectories intersect the current limit circle at point A_1 . The segment OA_1 within the current limit circle represents the region where the PMSM can operate at constant torque. The voltage limit ellipse passing through each point on this segment corresponds to the maximum achievable speed (transition speed ω_{tr}) of the PMSM. Higher constant torque values correspond to larger semi-axis lengths of the voltage limit ellipse and result in lower transition speeds. Where point A_1 corresponds to the maximum torque. Note that point A_1 relates to all operating points in the constant torque region below the base speed on the external characteristic curve (i.e., for $0 \leq \omega_r \leq \omega_{rb}$), including the segment between points B_0 and B_{rb} , as shown in Fig. 6.16.

When the speed of the PMSM reaches the base speed ω_{rb} at point A_1 , i.e., the PMSM operates at point $B_{rb}(\omega_{rb}, t_{e\max})$ on the external characteristic, the voltage limit circle becomes active. At this moment, according to Eqs. (6.35) and (6.36), the voltage limit equation is given by:

$$u_{Qm} = R_s i_Q + \omega_{rb} (L_D i_D + \psi_f) \quad (6.47)$$

Fig. 6.16 Constant torque and constant power operation (external characteristics) curves



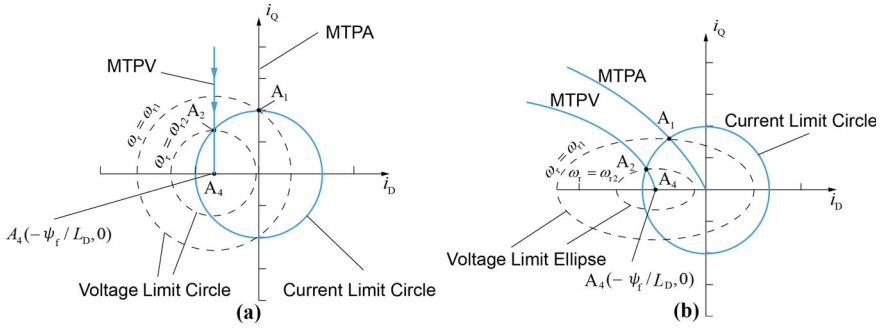


Fig. 6.17 Field-weakening control and optimal stator current control: **a** Surface-mounted; **b** interior-type

$$u_{Dm} = R_s i_D - \omega_{rb} L_Q i_Q \quad (6.48)$$

where u_{Qm} and u_{Dm} represent the peak values of the quadrature-axis and direct-axis components of the stator voltage $\mathbf{u}_s$, respectively. At this moment, the magnitude of the stator voltage reaches its maximum value U_{sm} .

For the operating point A_1 , its dynamic voltage equation can be derived from Eq. (6.17) as follows:

$$L_D \frac{di_D}{dt} = u_{Dm} - R_s i_D + \omega_{rb} L_Q i_Q = 0 \quad (6.49)$$

$$L_Q \frac{di_Q}{dt} = u_{Qm} - R_s i_Q - \omega_{rb} (L_D i_D + \psi_f) = 0 \quad (6.50)$$

It can be seen that when the PMSM operates at point B_{rb} , the current regulator has entered a saturated state, resulting in the control system losing its ability to regulate the stator current.

Under such conditions, the stator current vector $\mathbf{i}_s$ will deviate from point A_1 on the current phase plane. As shown in Fig. 6.17b, considering the voltage constraint, the operating point can only swing to the left (within the voltage ellipse). If the quadrature-axis component i_Q can be gradually reduced and the direct-axis component i_D increased negatively from point A_1 , it will force $\mathbf{i}_s$ to swing leftward. According to Fig. 6.17 and Eq. (6.37), this adjustment will reduce the stator voltage magnitude $|\mathbf{u}_s|$, making $|\mathbf{u}_s| < U_{sm}$, thereby allowing the regulator to exit saturation. The system can then regain control over the stator current. As i_D continues to increase (in the negative direction) and i_Q decreases, the rotor speed range can be gradually extended, the torque will deviate from its maximum value, and the PMSM will enter the constant power operating region.

This effect is primarily due to the demagnetizing effect on the permanent magnets caused by the negative direct-axis current's MMF, which weakens the direct-axis magnetic field. Therefore, this process is referred to as field-weakening. In the

field-weakening process, the control of i_D and i_Q is known as field-weakening control (Vaez-Zadeh 2018).

If the stator current magnitude is maintained at its rated value during field-weakening control, the trajectory of the stator current vector i_s will gradually move along the circumference from point A_1 toward point A_2 . When the control angle is with $\beta = 180^\circ$, i.e., $i_Q = 0$, the stator current consists entirely of D-axis demagnetizing current, thus:

$$\omega_{r\max} = \frac{U_{sm}}{\psi_f + L_D i_D} \quad (6.51)$$

An extreme case occurs when $\psi_f + L_D i_D = 0$, at which point the PMSM speed would theoretically increase to infinity. This operating point corresponds to the center of the voltage limit ellipse in Fig. 6.17, denoted as point A_4 , with the coordinate of $A_4(-\psi_f/L_D, 0)$. However, this condition rarely occurs in practice. To achieve $\psi_f + L_D i_D = 0$ during actual operation, it would require that $i_D = -\psi_f/L_D$. Since ψ_f/L_D is typically a large value, and $|i_D|$ is limited by the current limit circle, the field-weakening effect is therefore limited. Even if the inverter could provide a higher demagnetizing current, excessive demagnetization could cause irreversible demagnetization of the permanent magnets. Compared with three-phase induction motors, the field-weakening capability of PMSMs is limited, and the speed extension range is restricted, which is a technical shortcoming of PMSMs.

2. MTPV

The following discusses how to optimally control the stator current vector during field-weakening operation to meet the requirement of maximum power output.

In the field-weakening region, the motor typically operates at constant power output, but it may also be required to deliver maximum possible power. Since the motor output power is the product of torque and speed, achieving maximum output power can be interpreted in two ways: delivering the maximum torque at a given speed, or reaching the maximum speed under a given torque condition.

Therefore, under a fixed voltage constraint and constant speed, achieving a given torque with the minimum voltage corresponds to the so-called maximum torque per voltage (MTPV) control. From this perspective, the physical interpretation of MTPV is essentially equivalent to that of maximum power output control. On the $i_D - i_Q$ phase plane, this is represented by the trajectory connecting the tangential points between the constant torque curves and the voltage limit ellipses.

The analytical derivation of the MTPV condition is presented below:

The torque expression can be written in terms of flux linkages as

$$T_e = \frac{3}{2} p_0 (\psi_D i_Q - \psi_Q i_D) = \frac{3}{2} p_0 \left[\psi_D \frac{\psi_Q}{L_Q} - \psi_Q \frac{\psi_D - \psi_f}{L_D} \right] \quad (6.52)$$

The voltage limit ellipse can also be expressed in terms of flux linkage

$$\psi_D^2 + \psi_Q^2 \leq \left(\frac{U_{sm}}{\omega_r} \right)^2 \quad (6.53)$$

Expressing the flux linkage in polar coordinates, it holds

$$\begin{cases} \psi_D = \psi_s \cos \alpha \\ \psi_Q = \psi_s \sin \alpha \end{cases} \quad (6.54)$$

where ψ_s is the amplitude of the stator flux linkage, substituting Eq. (6.54) into the torque equation yields

$$t_e = \frac{3p_0}{2} \left[\left(\frac{1}{2L_Q} - \frac{1}{2L_D} \right) \psi_s^2 \sin 2\alpha + \frac{\psi_f}{L_D} \psi_s \sin \alpha \right] \quad (6.55)$$

Under voltage saturation conditions, the flux linkage satisfies

$$\psi_{sm} = \frac{U_{sm}}{\omega_r} \quad (6.56)$$

By maximizing the torque expression in Eq. (6.55), the coordinates of the intersection point in the flux linkage phase plane can be obtained.

$$\left. \frac{\partial t_e}{\partial \alpha} \right|_{\psi_s = \psi_{sm}} = \frac{3p_0}{2L_Q L_D} \left[(L_D - L_Q) \psi_{sm}^2 \cos 2\alpha + L_Q \psi_f \psi_{sm} \cos \alpha \right] = 0 \quad (6.57)$$

Then,

$$(L_D - L_Q)(\psi_D^2 - \psi_Q^2) + L_Q \psi_f \psi_D = 0 \quad (6.58)$$

Therefore,

$$\psi_Q = \pm \sqrt{\psi_D \left(\psi_D + \frac{L_Q}{L_D - L_Q} \psi_f \right)} \quad (6.59)$$

By rewriting Eq. (6.59) in terms of current variables, the analytical expression for MTPV trajectory is obtained

$$i_Q = \pm \sqrt{\frac{L_D(L_D i_D + \psi_f) \left[(L_D - L_Q) i_D + \psi_f \right]}{(L_D - L_Q) L_Q^2}}, \quad i_D < -\frac{\psi_f}{L_D} \quad (6.60)$$

The constant torque curve, current limit circle, voltage limit ellipse, MTPA trajectory, and MTPV trajectory can be shown on the current phase plane based on the above analysis, as shown in Fig. 6.18.

Figure 6.19 illustrates the trajectory of the stator current vector that enables maximum power output. This trajectory intersects the current limit circle at point A_2 , which corresponds to the speed ω_{r2} . This is the minimum speed at which the motor can deliver maximum power output under the voltage constraint. When the speed falls below ω_{r2} , the intersection point between the stator current trajectory and the voltage limit ellipse will lie outside the current limit circle. If $\psi_f/L_D > I_{sm}$, the maximum power output trajectory will entirely lie outside the current limit circle,

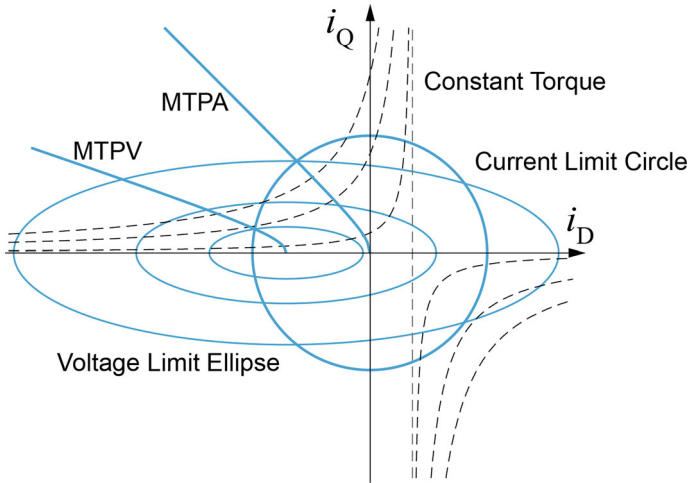


Fig. 6.18 The constant torque curve, current limit circle, voltage limit ellipse, MTPA trajectory, and MTPV trajectory of PMSM control

as shown in Fig. 6.19a. In this case, the maximum power output control trajectory becomes unattainable.

In summary, as shown in Fig. 6.19, the stator current vector can be controlled over the entire speed range on the external characteristic curve as follows:

- Interval I ($\omega_r \leq \omega_{r1}$): The stator current can be controlled according to Eq. (6.31), and the stator current vector will vary along the maximum torque/current ratio trajectory.
- Interval II ($\omega_{r1} < \omega_r \leq \omega_{r2}$): If the PMSM is operating at A_1 and the speed has reached the base speed ($\omega_{r1} = \omega_{rb}$), the stator current vector can be controlled to move leftward along the circumference from A_1 for field-weakening control. As the speed increases, the stator current vector moves from A_1 to A_2 .

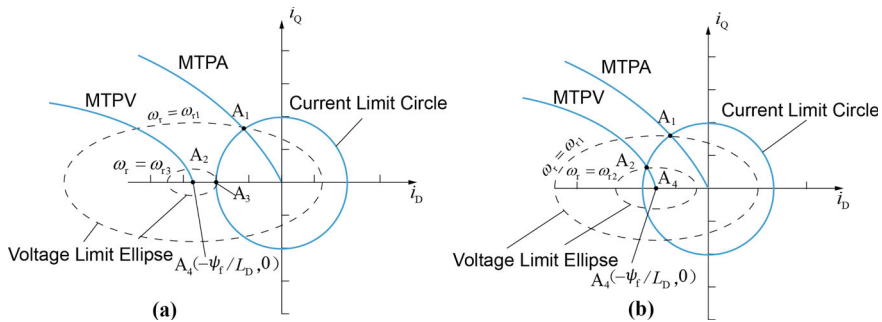


Fig. 6.19 Stator current vector trajectory: **a** $\psi_f/L_D > I_{sm}$; **b** $\psi_f/L_D < I_{sm}$

- Interval III ($\omega_r > \omega_{r2}$): i_D and i_Q can be controlled according to Eq. (6.60), with the stator current vector moving along the maximum power output trajectory from A_2 to A_4 . Of course, if $\psi_f/L_D > I_{sm}$, the control in Interval III does not exist. In this case, the control in Interval II can be extended from point A_2 to A_3 , as shown in Fig. 6.19. The speed corresponding to point A_3 is denoted as ω_{r3} , which represents the theoretically maximum achievable speed under field-weakening control.

The above discussion mainly focuses on control strategies along the external characteristic. For partial-load conditions, different control methods can be adopted depending on the current allocation strategy. These will be further discussed in Chap. 9 in the context of automotive operating conditions.

6.4.3 Basic Principle of PMSM Braking

Under braking conditions, the PMSM operates in the third and fourth quadrants, where the torque direction is opposite to the rotational speed direction, and the motor absorbs mechanical power. For interior-type PMSMs, a current allocation strategy based on the MTPA principle can be employed to ensure operation in the third quadrant.

To simplify the analysis, the following discussion takes a surface-mounted PMSM as an example, where the direct-axis current is controlled to zero with $i_D = 0$. The vector diagrams for different operating modes are shown in Fig. 6.20, where the counterclockwise direction of rotation is defined as positive.

The expressions for torque, mechanical power, and electrical input power are as follows:

$$t_e = \frac{3}{2} p_0 \psi_f \times i_s \quad (6.61)$$

$$P_m = t_e \omega_r \quad (6.62)$$

$$P_e = \frac{3}{2} \mathbf{u}_s \cdot \mathbf{i}_Q = \frac{3}{2} u_s i_Q \cos \varphi \quad (6.63)$$

Firstly, under driving mode (Fig. 6.20a), the motor controller outputs a voltage vector to overcome the back EMF and maintain the required (forward) drive current in the stator. The power factor angle between the stator voltage vector and current vector is with $\varphi < 90^\circ$. The rotor rotates counterclockwise at an angular speed of ω_r , and from the permanent magnet flux linkage and stator current vector, it is known that the current leads the magnetic field, with the torque angle $\beta = 90^\circ$, and the torque is counterclockwise with $t_e = 3p_0\psi_f i_Q/2$, thus the PMSM generates driving torque, and the power P_m is positive.

Under braking conditions (Fig. 6.20b), the stator current lags behind the rotor magnetic field, resulting in a torque angle of $\beta = -90^\circ$. The electromagnetic torque

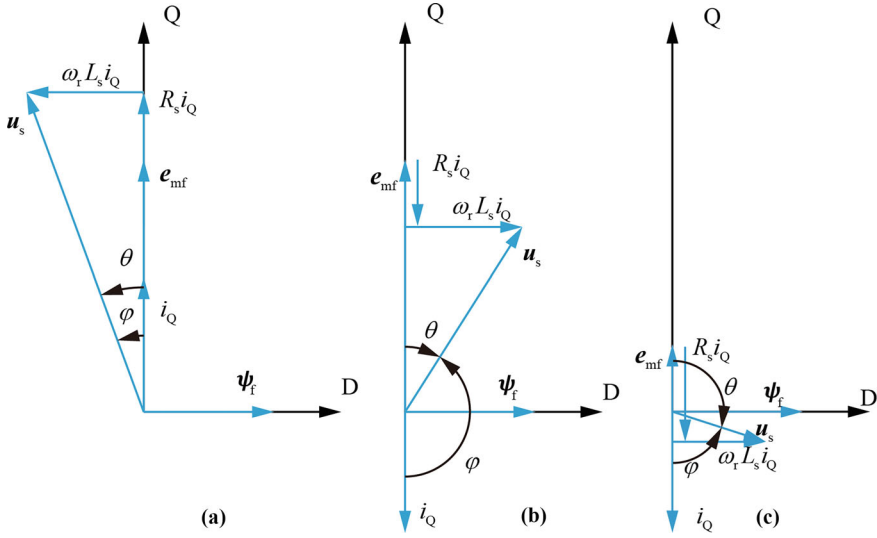


Fig. 6.20 Vector diagrams under FOC of a PMSM: **a** Driving mode; **b** regenerative braking; **c** energy-consuming braking

acts in the clockwise direction and is given by $t_e = 3p_0\psi_f i_Q/2$, $i_Q < 0$, indicating that the motor absorbs mechanical power. In this state, the power factor angle between the stator voltage vector and current vector is with $\varphi > 90^\circ$, resulting in negative electrical power, i.e., the PMSM feeds electrical power back to the power source. In this case, the back EMF of the PMSM is sufficient to sustain the required braking current against the source voltage. The angle between the stator current vector and voltage vector is greater than 90° , enabling energy to flow back into the battery. The motor operates in generating mode, and this condition is referred to as **regenerative braking**.

When the PMSM speed is relatively low and the back EMF of the PMSM e_{mf} is insufficient to overcome the supply voltage and maintain the required braking current, in order to ensure the desired braking torque, the stator voltage vector must be controlled into the fourth quadrant so that the angle between the voltage and current vectors is less than 90° . In this case, the battery must supply part of the electrical power to sustain the braking current. The PMSM operates in an **energy-consuming braking** mode, where both the generated power and the power drawn from the battery are dissipated as losses in the phase resistance.

From the stator voltage vector equation, it has

$$\mathbf{u}_s = R_s i_Q + j\omega_r L_s i_Q + j\mathbf{e}_{mf} \quad (6.64)$$

when the speed is relatively high, and the current i_Q satisfies $R_s i_Q < e_{mf}$, regenerative braking occurs; conversely, in relatively low-speed and high-torque braking conditions, if $R_s i_Q > e_{mf}$, energy-consuming braking occurs.

6.5 Summary

The rotor structure of a PMSM can be categorized into two main types based on the mounting configuration of the permanent magnets: surface-mounted and interior-type. In an interior-type PMSM, the permanent magnets are embedded or buried within the rotor core, resulting in a non-uniform air gap. This forms a salient-pole rotor structure, which introduces additional reluctance torque. The presence of reluctance torque enhances the torque-per-ampere ratio, facilitating field-weakening operation and extending the speed range of the PMSM. As a result, interior-type PMSMs are predominantly used in automotive applications.

Based on the four-coil primitive machine introduced in Chap. 4, this chapter derives the flux linkage equations, voltage equations, and torque expression for the PMSM, which serve as the fundamental basis for subsequent analysis and control design.

Rotor magnetic field-oriented FOC is the primary control strategy for PMSMs. The optimal control of stator current involves selecting an appropriate current combination from the constant torque curves that yields a desired torque. In the low-speed operating region, the stator current is typically regulated based on the MTPA principle, which minimizes copper losses in the motor and reduces losses in the motor controller.

The maximum voltage that the motor controller can supply to the PMSM is constrained by the available DC voltage from the onboard power source. The magnitude of the stator voltage vector $\mathbf{u}_s$ required for steady-state operation is directly proportional to the rotor electrical angular speed ω_r , implying that the operating speed of the PMSM is limited by the inverter's voltage capability. To represent this constraint, a family of voltage limit ellipses is introduced. The higher the motor speed, the smaller the corresponding ellipse. When the speed exceeds the base speed, field-weakening control is required to overcome the limitations imposed by the voltage constraint.

Exercises

- 6.1 The instantaneous values of the three-phase stator current of the PMSM are $i_A = -1\text{A}$, $i_B = 2\text{A}$, $i_C = -1\text{A}$. Please represent the magnitude and phase of i_s at this moment on the space complex plane with the A-axis as the real axis using a vector diagram.
- 6.2 Please discuss the principle of torque generation in PMSM. In conjunction with the surface-mounted PMSM stator magnetic field vector equation $\psi_s = L_s \psi_s + \psi_f$, explain the three (spatial sinusoidal distribution) magnetic fields present in PMSM and why the electromagnetic torque can be expressed by the interaction of any two of these magnetic fields. Try to derive the corresponding electromagnetic torque vector equations.
- 6.3 Please explain why rotor field-oriented vector control can make the PMSM equivalent to a separately excited DC motor.

- 6.4 Please explain why interior-type PMSM generates reluctance torque and how reluctance torque is controlled in rotor field-oriented vector control.
- 6.5 Try to prove that under sinusoidal steady-state and equal amplitude coordination transformation conditions, the magnitude of the current space vector is I_s , where I_s is the peak value of the phase current.
- 6.6 If the magnetic field in the motor air gap is not sinusoidally distributed (or does not take the fundamental wave), can space vector theory still be used to analyze the motor? Why?
- 6.7 The equivalent physical model of the PMSM is shown in Fig. 6.21. Please provide:
- (1) The expression for mutual inductance between the two coils;
 - (2) The expression for the magnetic co-energy of the coils;
 - (3) Derive the torque expression based on the relationship between magnetic co-energy and torque.
- 6.8 The physical model of a single-coil excited motor is shown in Fig. 6.22 ($L_Q > L_D$). Please provide:
- (1) The expression for the self-inductance of the coil as a function of angle;
 - (2) The expression for the magnetic co-energy of coil A;
 - (3) Derive the reluctance torque expression.
- 6.9 The equivalent physical model of a salient-pole motor is shown in Fig. 6.23. Please provide:
- (1) The expression for the stator self-inductance as a function of angle β ;
 - (2) The expression for mutual inductance between the stator coil and rotor coil;
 - (3) Derive the expression for the magnetic co-energy of the equivalent model;
 - (4) The torque expression.

Fig. 6.21 The equivalent physical model of the PMSM for Exercise 6.7

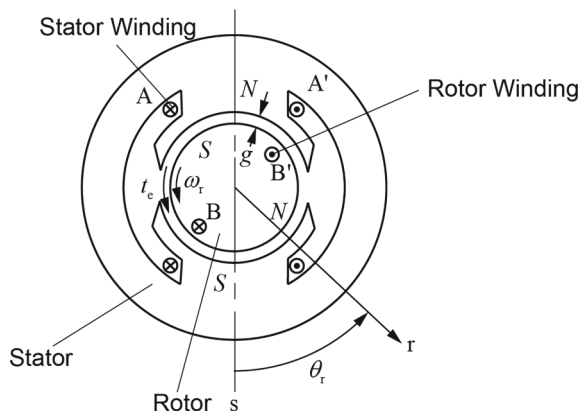


Fig. 6.22 The physical model of a single-coil excited motor for Exercise 6.8

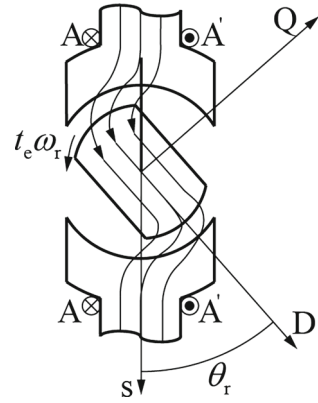
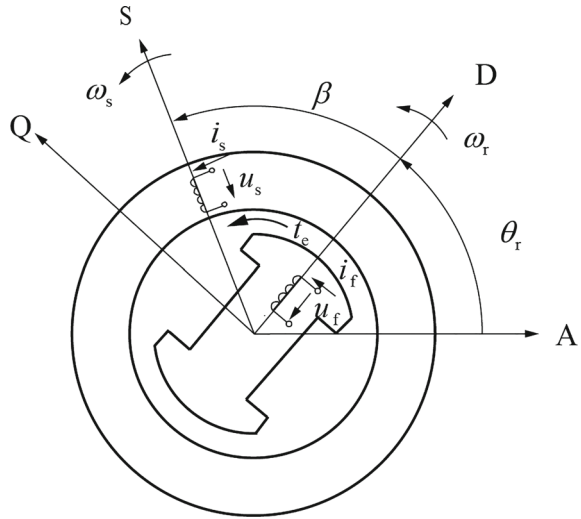


Fig. 6.23 The equivalent physical model of a salient-pole motor for Exercise 6.9

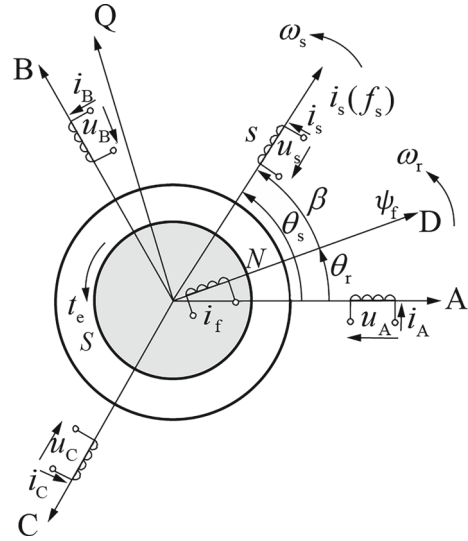


6.10 The equivalent physical model of a surface-mounted three-phase PMSM is shown in Fig. 6.24. Based on amplitude equivalence, the stator current vector expression is $i_s = 2/3(i_A + ai_B + a^2i_C)$. Please provide:

- (1) The expression for the stator flux linkage vector;
- (2) The expression for the permanent magnet flux linkage and its derivative;
- (3) The expression for the stator voltage vector;
- (4) Further write the voltage vector equation under sinusoidal steady-state current excitation and draw the steady-state vector diagram.

6.11 The dual-axis equivalent physical model of an interior-type three-phase PMSM is shown in Fig. 6.25. In the DQ coordinate system, the expressions for current, flux linkage, and voltage are: $i_s^D = i_D + ji_Q$, $\psi_s^D = \psi_D + j\psi_Q = L_D i_D +$

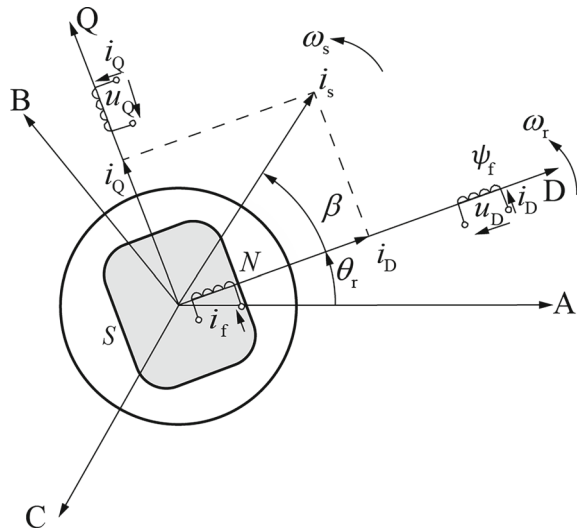
Fig. 6.24 The equivalent physical model of a surface-mounted three-phase PMSM for Exercise 6.10



$\psi_f + jL_Q i_Q, u_s^D = u_D + ju_Q$. Try to derive the voltage equation in the DQ coordinate system, draw the vector diagram, and write the torque equation in DQ coordinates.

- 6.12 Discuss the basic principles of field-weakening control for interior-type PMSMs.
- 6.13 Please write the current expression for the MTPA control of a surface-mounted PMSM.
- 6.14 The known PMSM parameters are as follows:

Fig. 6.25 The equivalent physical model of an interior-type three-phase PMSM for Exercise 6.11



- (i) 0.88 kW PMSM: Stator phase resistance $R_s = 4m\Omega$, peak value of the permanent magnet flux linkage of the stator winding $\psi_f = 0.055\text{Wb}$, number of pole pairs $p_0 = 4$, moment of inertia $J = 0.048\text{ kgm}^2$, stator winding phase leakage inductance $L_\sigma = 20\mu\text{H}$, D-axis excitation inductance $L_D = 88\mu\text{H}$, Q-axis excitation inductance $L_Q = 300\mu\text{H}$, base speed is 1676.2 rad/s , peak torque is $t_{\text{emax}} = 210\text{N} \cdot \text{m}$.
- (ii) 0.126 kW PMSM: Stator phase resistance $R_s = 8.3m\Omega$, peak value of the permanent magnet flux linkage of the stator winding $\psi_f = 0.071\text{ Wb}$, number of pole pairs $p = 4$, moment of inertia $J = 0.1\text{ kgm}^2$, stator winding phase leakage inductance $L_\sigma = 30\mu\text{H}$, D-axis excitation inductance $L_D = 174\mu\text{H}$, Q-axis excitation inductance $L_Q = 293\mu\text{H}$, base speed is 1968.75 rad/s , peak torque is $t_{\text{emax}} = 256\text{ N} \cdot \text{m}$.

Requirements: Use MATLAB for steady-state analysis of the PMSM, solve and/or plot.

- (1) Peak phase current corresponding to peak torque;
- (2) Constant torque curve (including asymptote);
- (3) Voltage limit ellipse (cluster);
- (4) MTPA curve;
- (5) MTPV curve;
- (6) Mechanical output characteristics (external characteristics);
- (7) Space vector diagram at base speed and maximum torque operating point (or other specific operating points);
- (8) Space vector diagram for base speed and maximum braking torque regenerative braking;
- (9) Optimal current distribution curve corresponding to MTPA;
- (10) “Torque angle characteristics” curves at different current amplitudes (plotting excitation torque, reluctance torque, and composite torque separately).

References

- Vaez-Zadeh S (2018) Control of permanent magnet synchronous motors. Oxford University Press, Oxford
- Wang C, Dong T, Peng B, Xia J (2022) Electric machinery. Mechanical Industry Press, Beijing

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